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1 Introduction

In this chapter, you will learn facts about the growth and development of countries in the world. First, you will see why the real per capita GDP can be used as an indicator of a country standard of living, and why economic growth can be a way of improving the well-being of individuals. Then, you will analyze the evolution of economic growth across countries and how it correlates with countries' characteristics. In particular, you will see that some poor countries have remained poor for decades while others have grown richer. Explaining this fact and finding solutions is one of the most important challenge that economists are facing. As a starting point, this chapter offers some clues by looking at comovements between growth and some variables and by comparing the characteristics of successful versus less successful countries.

Learning Outcomes

Students will be able to

- explain why real per capita GDP is used as an indicator of a country standard of living.
- interpret a distribution measured by bar and smooth densities.
- describe the evolution of the standard of living in the world.
- explain the importance of economic growth in the evolution of the standard of living.
- evaluate the level of convergence and conditional convergence among countries.
- identify variables that correlates with economic growth.
- identify the important issues that growth and development economists need to address.

Key Terms

- **Bar density:** This is a way of representing the distribution of variables. In a bar density, the area of each bar represents the proportion of observations that lie in the interval determined by its base. It is an alternative to histograms for which the height of the bars represents the number of observations.
- **Smooth density:** This is a line chart representing the distribution of a variable. It is approximately obtained by drawing a smooth line over a bar density. This tool is particularly useful to visualize multiple distributions on the same graph.
- **Economic growth:** This is the growth rate of the standard of living measured by the real per capita GDP over a long period of time.
- **Weighted distribution:** When we represent income distribution across countries using histograms, each country counts as one observation. To get a better representation of the distribution at the individual level, we need to take into account that the size of the population is not the same in each country. A weighted distribution will increase the importance of countries with larger population.
- **Convergence:** This is a mechanism through which the standard of living of countries measured by the real per capita GDP converges to the same value over a long period of time. We observe convergence when countries with lower standard of living grow faster.
- **Conditional convergence:** This is a mechanism through which the standard of living of similar countries measured by the real per capita GDP converges to the same value. We observe conditional convergence when countries with low standard of living grow faster than similar countries with high standard of living. Being similar could mean having an equally educated or healthy population, having a similar political system, having an equivalent respect of human rights, and so on.
- **Standard of living:** This is a subjective concept that characterizes the well being of individuals. The determinants of the standard of living may include, health, education, quality of air, freedom, security, income, etc. The real per capita GDP is often used as an indicator of the standard of living because it correlated with its determinants.
- **Correlation:** This term is a synonym for comovement or observed relationship. A variable correlates with another if there is a positive or negative comovement between the two variables. To indicate the direction of the comovement, we say positive correlation when the variables move in the same direction and negative correlation if they move in opposite directions.
- **Welfare:** This is a subjective concept that characterizes the well being of a country at the aggregate level. Welfare depends on the standard of living of each individual and on how it is distributed in the population. For example, if the standard of living of some individuals increases and it decreases for others, it is not clear what happens to the welfare of the economy.
- **Human development index (HDI):** This is an alternative measure of the aggregate standard of living. The index is based on education, life expectancy, the Gini coefficient and the real per capita income.
- **Inequality adjusted HDI (IHDI):** This is an alternative measure of the aggregate standard of living. The index is a modified version of the human development index HDI. It is constructed by taking into account inequality in education and health.

2 Economic Growth

Economic growth and development is an area of economics that studies the long term behaviour of economic activity and its implications on the well being of the population. In particular, development economists are interested in understanding why some countries are poor compared with others and why some countries who were poor 50 years ago are now among the richest while others remained poor. The following short videos from the United Nation Development Program (UNDP) explain well the issues that development economists are trying to address:

- Human Development Report 2019 - Animated Explainer (2:29 min)
- Animated video- Human Development Report 2016 (4:05 min)
- A New Generation of Inequality (1:41 min)

This chapter is therefore related to the chapter on income inequality. But, instead of studying the income inequality across individuals in a given economy, we are interested in analyzing inequality across countries and its evolution over time. In the chapters on unemployment and income inequality, we learned that whatever we are trying to analyze, we cannot rely on a single measure because the reality is more complex. In this chapter, we will be focusing on the long term behaviour of real per capita GDP, but we have to keep in mind that a understanding inequality and poverty requires an analysis at a more disaggregated level (at the individual or small group level). This is the message you should retain from the short videos.

2.1 Real Per Capita GDP Versus Welfare

This chapter focuses on the long term evolution of the standard of living across countries measured by the annual real per capita GDP. This is a measure of how much goods and services individuals can purchased on average over a year. We already know that the real per capita GDP is not allocated equally across individuals and therefore hides the existence of inequality. In this section, we want to address another possible weakness of this measure: its ability to measure the average standard of living or the average well being. In other words, can we measure the welfare of a group by how much goods and services they can purchased? This is a difficult question to answer because there is no objective definition of what we mean by welfare. The best we can do is to select variables that are likely to impact individuals' well being and see how they relate to real per capita GDP.

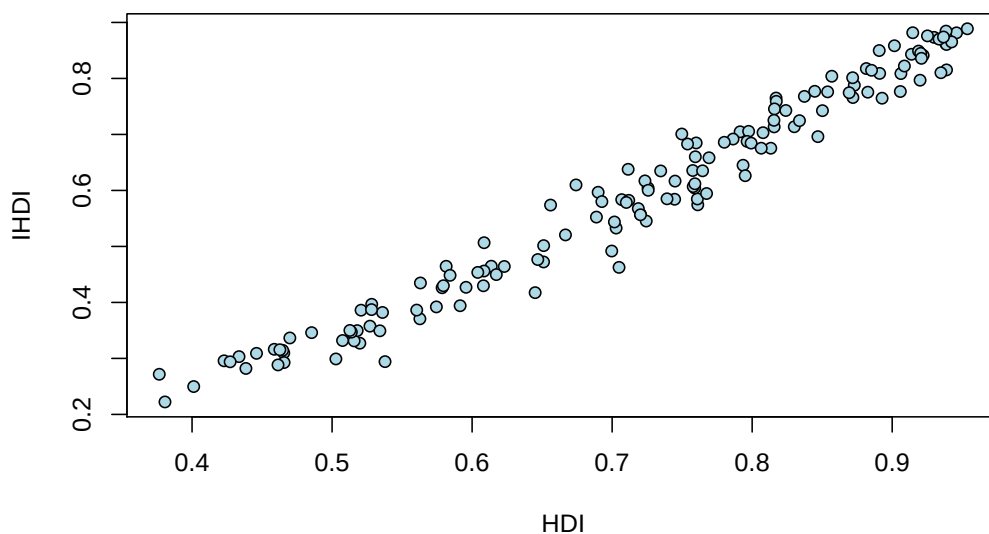
Human Development Index

The United Nation Development Program (UNDP) works with about 170 countries to help them reduce poverty and inequality. If you navigate through their sustainable development goals (SDGs), you will see that increasing real per capita GDP (economic growth) is just one variable among many. This list is a good starting point in understanding what determines welfare.

The UNDP publishes the human development index (HDI) in an attempt to better measure welfare. It includes the real per capita GDP, a measure of the average health quality (life expectancy) and a measure of the average level of education. Since 2010, they also publish an inequality adjusted index (IHDI) which includes inequality as a fourth component. The different variables used in this section are included in the The data file IHDI2018.csv.

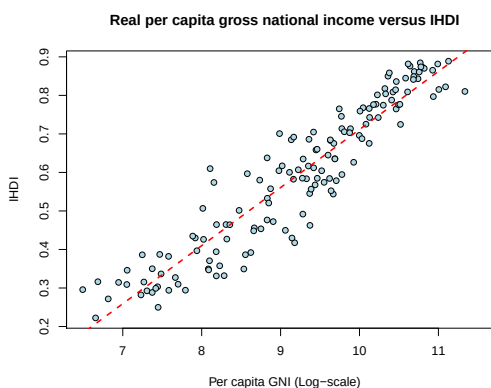
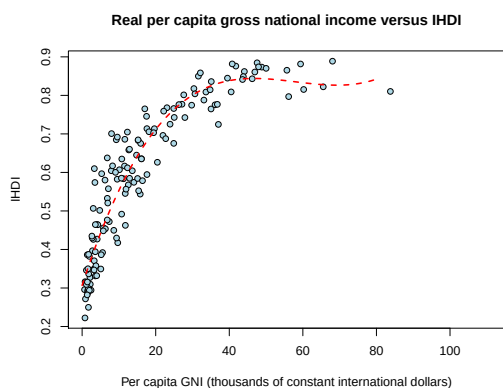
The following scatter plot shows that the two development indexes are closely related. We do see some variations (different IHDI for the HDI), but on average the two indexes are comparable.

IHDI versus HDI in 2018

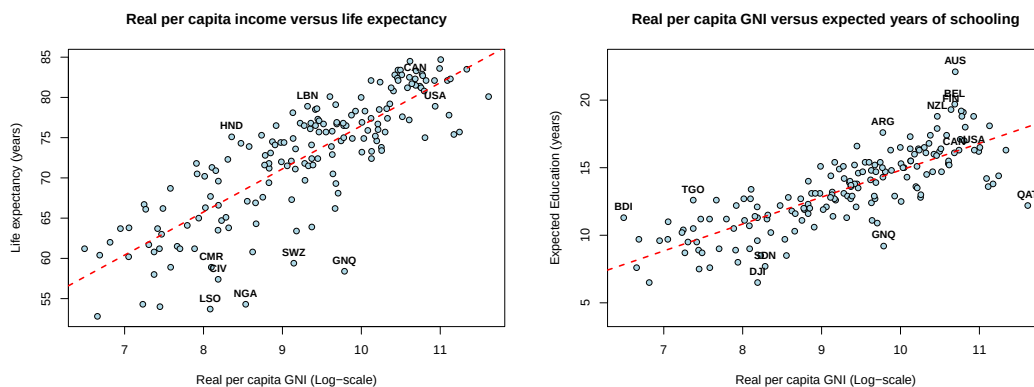


We want to see if the real per capita GDP is a good measure of welfare. The following is the scatter plot of IHDI as a function of GNI. On the left graph, GNI is expressed in thousands of constant international dollars and on the right graph GNI is expressed in logs. The red dashed lines have been added to show the overall comovement between the two variables. Clearly, both graphs show that there is a strong comovement between GNI and the human development index. However, there is some variations: we see IHDI gaps being above 0.3 for some countries with the same real per capita GNI. But, the variation decreases as GNI increases: we see little variation for countries with log GNI above 10.

The right graph was added to show you that the observed relationship is linear on average when GNI is expressed in logs. It means that a 1% increase in GNI implies a constant increase in IHDI on average (on the graph below each 1% increase of GNI is associated with IHDI being 0.0015 higher on average). Since is easier to interpret changes in percentage rather than changes in constant international dollars, we will only use that scale for analyze relationships.



Next, we want to see if there is a positive comovement between per capita income and other determinants of welfare. The following graphs are the scatter plots of life expectancy (left graph) and education (right graph) with respect to the log of real per capita GNI. Again, there is a strong positive comovement between real per capita income and the two variables. However, we also see lots of variations. For example, the expectancy for Lesotho (LSO) and Honduras (HND) are respectively 54 and 75 years despite the fact that they have similar real per capita income. We see the same gap between relatively richer countries such as Lebanon (LBN) and the Kingdom of Eswatini (SWZ). The same type of variations is observed if we use the expected level of education as welfare component (compare South Sudan (SDN) with Togo (TGO) or the Equatorial Guinea (GNQ) with Argentina (ARG)).



Concluding Remark

For the rest of the chapter, we will use real per capita GDP as an indicator of welfare. We are not by any mean pretending that real per capita GDP is a good measure of welfare. All we say is that it is a relatively good indicator because it is positively related to most components that determine welfare (education, health, reduced inequality, clean water and air, gender equality, peace and justice, etc.). We just have to keep in mind that higher real per capita GDP does not necessary imply welfare improvement: we just saw examples of richer countries with poor human conditions. What we do observe is that welfare tends to improve eventually as real per capita income increases.

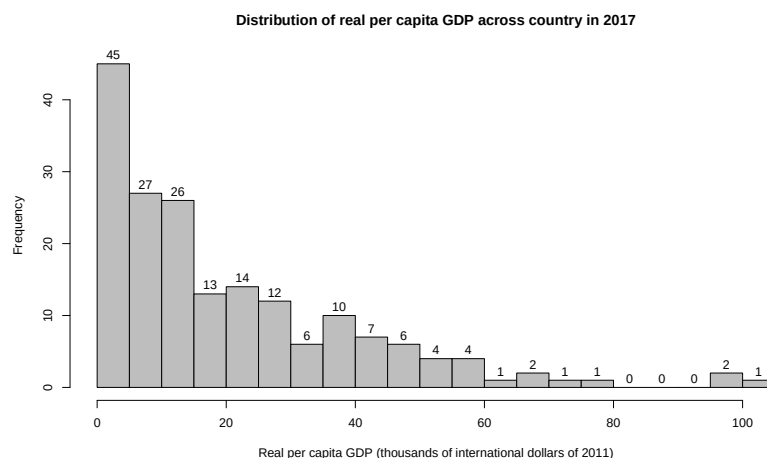
2.2 Inequality Across Countries

The data used in this section are from the Penn World Table (PWT) version 9.1 (Feenstra et al., 2015). The table is a collection of annual time series from 1950 to 2017 for 182 countries. Estimating macroeconomic indicators that can be compared across countries is a difficult tasks which explains why there are so many versions. Values in version 9.1 may also differ from order data sources such as the World Bank or the OECD web site. It is an issue for empirical macroeconomists trying to perform sophisticated statistical studies, but it has little impact on our graphical analysis. The PWT is useful because it contains longer time series than the World Bank or the OECD. However, the series are not complete for all countries. For example, the real GDP is available for only 30% of the 182 countries in 1950, 61% in 1960 and 86% in 1970.

Income Inequality

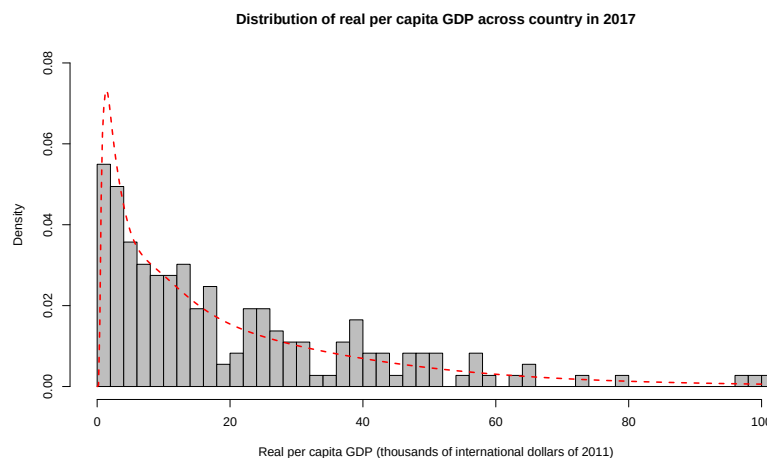
As we saw in previous chapter, total income is not allocated equally across countries. We start this section by reassessing this fact using world data. The following is the histogram of real per

capita GDP in 2017 for 182 countries (you can construct the series using the files PWT91pop.csv and PWT91rgdp.csv). We can see a large proportion of the countries at the bottom of the income distribution. In fact, more than half of the countries (98 out of 182) are represented by the first three bars, implying a per capita real GDP less than 15 thousand (the base of each rectangle is 5). The three richest countries at the right of the histogram are Luxembourg (99.5 thousand), Macao (97 thousand) and Qatar (100.1 thousand).



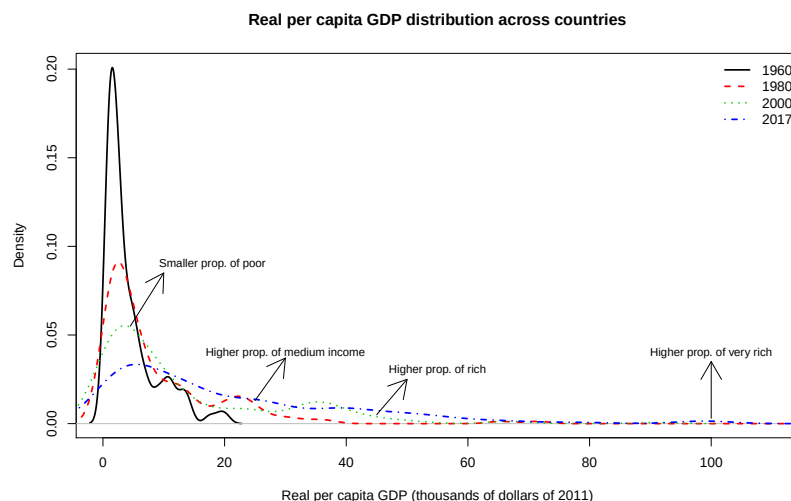
Distribution Using Smooth Density

When we want to compare income distributions over time, it is better to use smooth density because line charts are easier to interpret than bar charts when multiple series are represented. A smooth density is simply a smooth line that covers the bars of the histogram (it is more complicated than that, but it is good enough for this course). On the following histogram, the smooth density is represented by the red dashed line. You can see that the number of rectangles is much higher than in the previous graphs, because a smooth density is obtained by increasing the number of intervals. For this course, you don't need to know more about densities and you don't need to know how to create them. What you need to know is how to interpret them, which is identical to interpreting histograms.



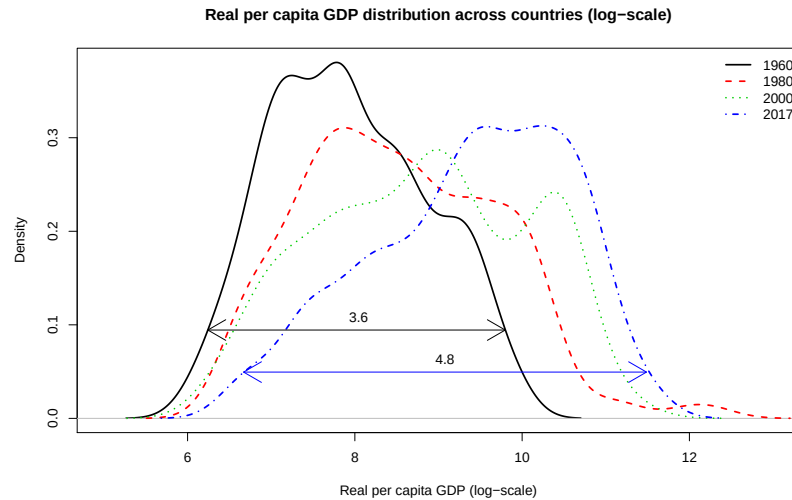
The Evolution of the Income Distribution Across Countries

The distributions for 1960, 1980, 2000 and 2017 are represented respectively by the solid black, the red dashed, the green dotted and the dot-dashed blue lines. Since the areas under the curves are proportions, we see that the proportion of poor countries has decreased during the periods 1960-1980, 1980-2000 and 2000-2017. Also, the growing density in the middle shows that the proportion of rich (around 40) and medium income (around 20) countries have increased over the period. Therefore, real per capita GDP has grown at different rates for different countries and some countries that were poor in the '60s are still poor today. As a result, we observe more inequality today than in the '60s. One of the purpose of economic growth is to explain this change in the income distribution.



Alternative Representation: Log Per Capita GDP Distribution

In the assigned readings, the author looks at the distribution of real per capita GDP expressed in logs. The following is what the author presents with the year 2017 added. You may see small differences because our graph is based on a different PWT version and also because the methods for generating the smooth densities may differ from mine. The distribution for 1960, 1980, 2000 and 2017 are represented respectively by the solid black, the red dashed, the green dotted and the dot-dashed blue lines.



We can see that the distribution is not as skewed as the previous one. However, we still observe that the gap between poor and rich countries have increased between 1960 and 2017. The arrows show the gaps between the first and last percentiles of the distribution in 1960 and 2017. The numbers look similar but they hide huge gaps. In 1960, the gap in log is 3.6, which means:

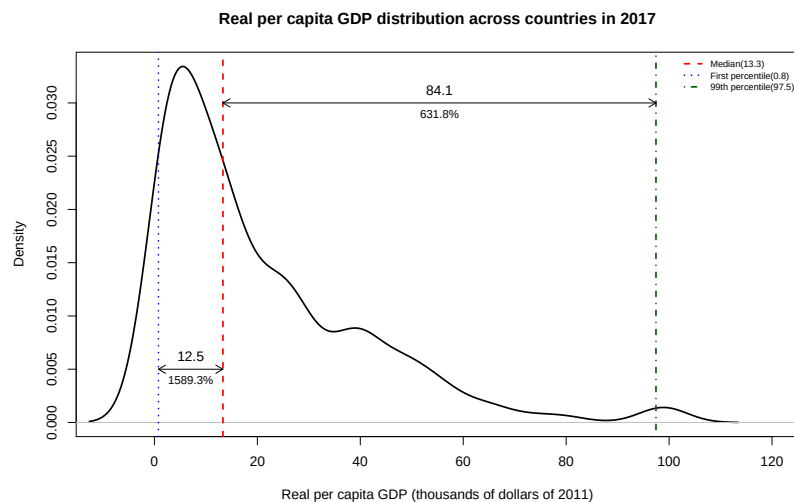
$$\begin{aligned}\log\left(\frac{X_{99}}{X_1}\right) &= 3.6 \\ \frac{X_{99}}{X_1} &= e^{3.6} = 36.6,\end{aligned}$$

where X_1 and X_{99} are the first and 99th percentiles. It means that the per capita income of the 99th percentile is 36.6 times higher than the per capita income of the first percentile. In 2017, the per capita income of the 99th percentile is 121.5 higher:

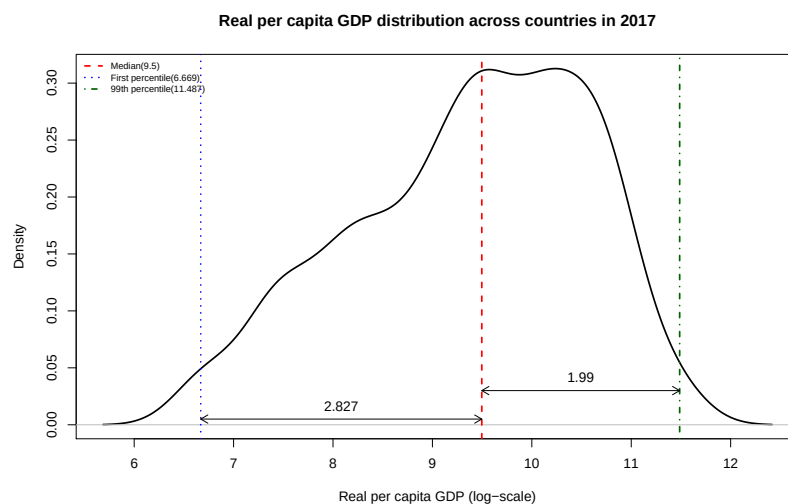
$$\begin{aligned}\log\left(\frac{X_{99}}{X_1}\right) &= 4.8 \\ \frac{X_{99}}{X_1} &= e^{4.8} = 121.5.\end{aligned}$$

Therefore, the shape of the distributions in dollars versus in logs are different because large differences in dollars translates into small differences in logs. This is why the the distribution looks more equally spread on both sides of the center when the variable is expressed in logs.

To understand why the distribution of the log is less skewed, the following shows the distribution of real per capita GDP expressed in thousands of dollars for 2011. On the graph, the red dash line represents the median, and the dotted blue and dot-dashed green lines are respectively the per capital incomes of the first and 99th percentiles. From the numbers on top of the arrows, we see that the average per capita income of the 99th percentile is 84.1 thousand dollars higher than the median and it is 12.5 thousand dollars lower for the first percentile. This is a common characteristic for all skewed distributions (longer right tail). In proportion, however, the gap is reversed (numbers under the arrows). The median is 1589.3% higher than the first percentile and the 99th percentile is 631.8% higher than the median.



The difference in proportion is what we measure when we use the log scale. The following graph is the distribution using the log scale. As for the previous graph, the red dash line represents the median, and the dotted blue and dot-dashed green lines are the per capital incomes of the first and 99th percentiles.



Using the same log formula we obtain approximately the same percentage differences (X_m is the median and the difference of percentage gaps are due to rounding errors and how the percentiles are computed in the two cases):

$$\begin{aligned} \log(X_m) - \log(X_1) &\equiv \log\left(\frac{X_m}{X_1}\right) = 2.827 \\ \frac{X_m}{X_1} &= e^{2.827} = 16.8947 \\ \frac{X_m - X_1}{X_1} \times 100\% &= [16.8947 - 1] \times 100\% = 1589.47\% \end{aligned}$$

and

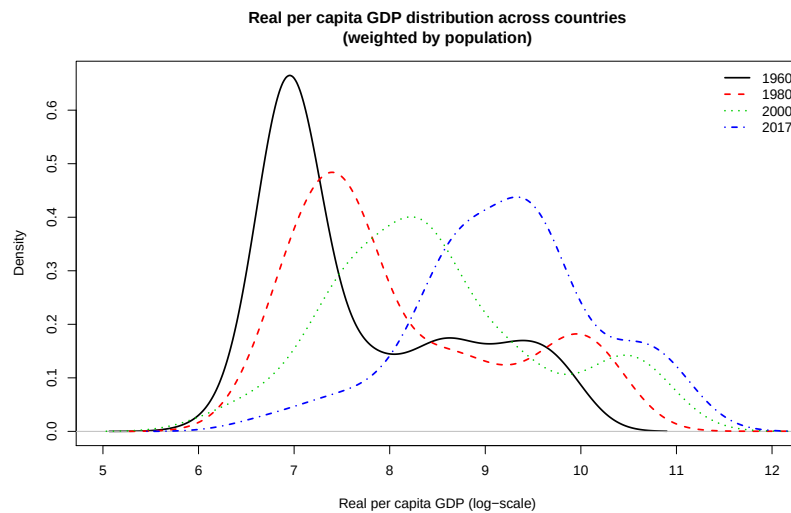
$$\begin{aligned}\log(X_{99}) - \log(X_m) &\equiv \log\left(\frac{X_{99}}{X_m}\right) = 1.99 \\ \frac{X_{99}}{X_m} &= e^{1.99} = 7.3155 \\ \frac{X_{99} - X_m}{X_m} \times 100\% &= [7.3155 - 1] \times 100\% = 631.55\%.\end{aligned}$$

Therefore, we need to dig a little deeper to understand distributions of logs. The most important result to remember is that despite the fact that the distribution is less skewed, we still observe an increase in the gap between rich and poor countries.

Alternative Representation: Weighed Distribution

The author also presents the weighted distribution. The problem with the above method is that it does not take into account that some countries are much more populated than others. For example, the population of China in 2017 was 1.4 billion and the population of Montserrat was 5.2 thousand. Clearly, we cannot consider an increase of the real per capita GDP in China equivalent to an increase in Montserrat: the former implies that 1.4 billion individuals are richer on average while the latter implies only 5.2 thousand richer individuals on average. Therefore, if we want to visualize the income distribution in the world at the individual level, we need to increase the importance of populated countries relative less populated ones. Learning how to construct a weighted distribution is beyond the scope of this course but intuitively, it is like constructing a distribution from a series in which the real per capita GDP of China is repeated 1.4 billion times, the real per capita GDP of Montserrat is repeated 5.2 thousand times and so on.

The following graphs reproduces Figure 1.3 of the assigned readings with the year 2017 added. The distribution for 1960, 1980, 2000 and 2017 are represented respectively by the solid black, the red dashed, the dotted green and the dot-dashed blue lines. We can see that the per capita income as increased for a large proportion of the world population between 1960 and 2017 (large shift of the hill to the right). This is partly due to the economic growth in China and India, two countries that represent about 30% of the world population (real per capita GDP was multiplied by 10 in China between 1960 and 2017 and it was multiplied by 4 in India). However, the gap between the poor and rich has increased even at the individual level (a difference in logs of 3.5 in 1960 and 4.3 in 2017 between percentiles 1 and 99, not shown on the graph).



2.3 The Importance of Growth

This chapter is called Economic growth (growth rate of the real per capita GDP) because growth is what allows poor countries to become richer. In fact, a small difference in the annual growth rate over a long period of time has a huge impact on the standard of living. To see that, suppose that the initial real per capita GDP is X_0 and the annual growth rate is g for T years. We saw in a previous chapter that it implies the following real per capita GDP after T years:

$$X_T = (1 + g)^T X_0.$$

Therefore, the real per capita GDP is multiplied by $(1 + g)^T$. For example, suppose two countries start with the same real per capita GDP, but the annual growth rate over a period of 50 years in the first country is 1% and it is 2% in the second. Then, the real per capita GDP is multiplied by

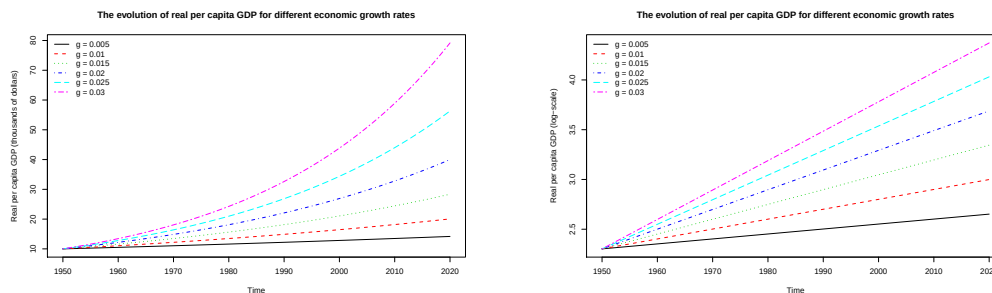
$$(1 + g)^T = (1.01)^{50} = 1.6$$

in the first country and by

$$(1 + g)^T = (1.02)^{50} = 2.7$$

in the second. For example, if X_0 was 10 thousand dollars for both countries, the real per capita GDP after 50 years would become 16 thousand dollars in the first country and 27 thousand in the second. Therefore, the extra percentage point results in 11 thousand dollars more real income by individual on average.

The following graph compares the trending behaviour of real per capita GDP for countries with the same initial value but with annual growth rates ranging from 0.5% to 3%, over a period of 70 years (1950 to 2020). On the left graph the GDP is expressed in thousands of dollars and it is expressed in logs on the right graph.

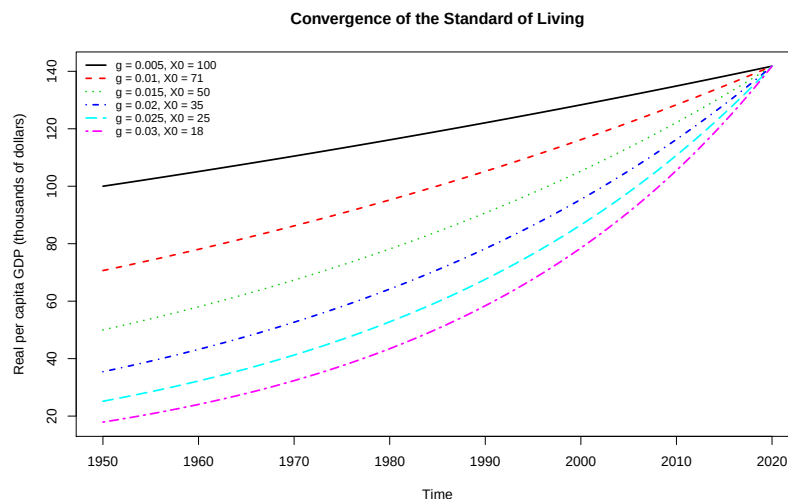


We can see that gaps between slow and fast growing economies are increasing over the period. Therefore, inequality in economic growth rates leads to income inequality. Notice that the gap in dollars between consecutive trends in a given year increases as we move up on the left graph. In other words, the difference of GDP in dollars in 2020 between the countries with $g = 0.005$ and $g = 0.01$ is smaller than the difference between the countries with $g = 0.025$ and $g = 0.03$. However, the right graph shows that the percentage differences are the same (same difference in logs implies same percentage difference).

2.4 Convergence and Conditional Convergence

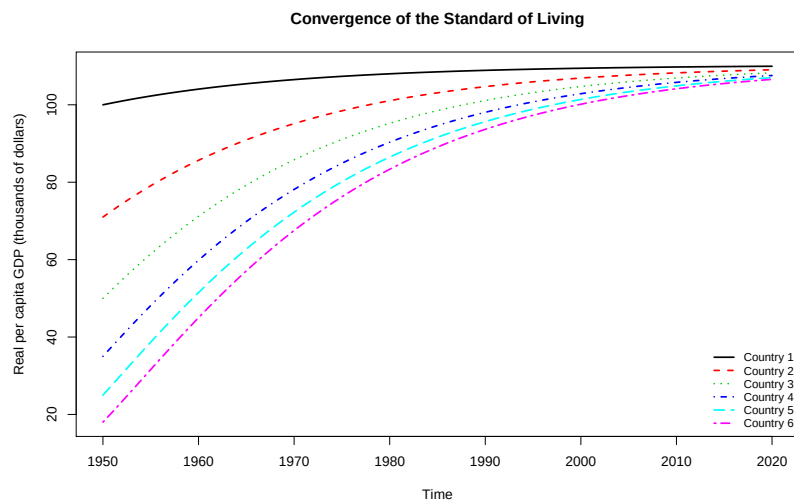
The Concept of Convergence

Some models from the economic growth theory predicts that inequality across countries (or some countries) should eventually be reduced, because poor countries should grow faster than rich countries. This result is called convergence because it implies that the standard of living of poor and rich countries will eventually converge to the same. The following graph illustrates the concept of convergence. The growth rates are the same used in the previous graphs but the 1950 per capita GDP of the faster growing countries are set to a lower value. Since poor countries grow faster (e.g. the dot-dashed pink line), they eventually reach the same standard of living as the countries that were initially rich (e.g. the solid black line).



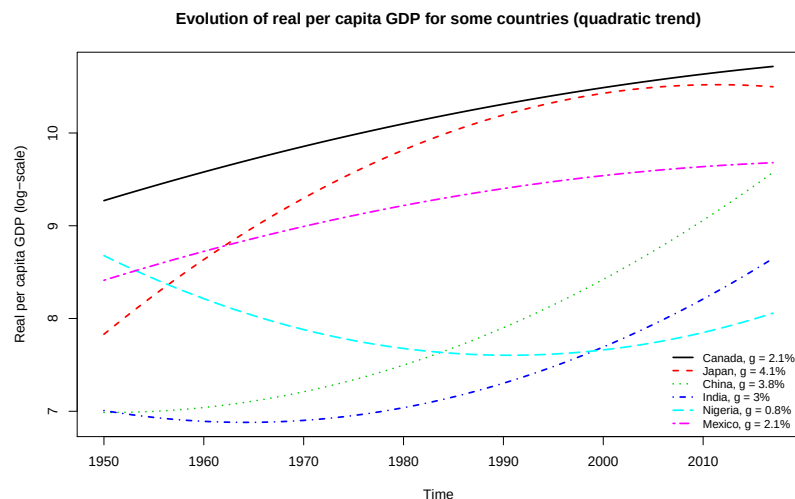
The above graph does not show exactly what convergence means in theory. If poor countries grow faster than rich countries, then their growth rates should decrease as they get richer, which is not what

we observe on the graph. In fact, if we continued the lines after 2020, we would observe a divergence with the poorest country becoming the richest and the richest becoming the poorest. The purpose of the graph is just to explain what it means for countries to converge to the same real per capita GDP. We also want to point out that convergence exists only if poor countries grow faster than rich countries. The following graph shows what convergence means: the growth rate is slowing down as countries get richer. If we were to continue the lines passed 2020, all lines would get closer and closer from each other. It implies that at any point in time poorer countries grow faster. That's the only fact about convergence that we need to understand before going to the next section.



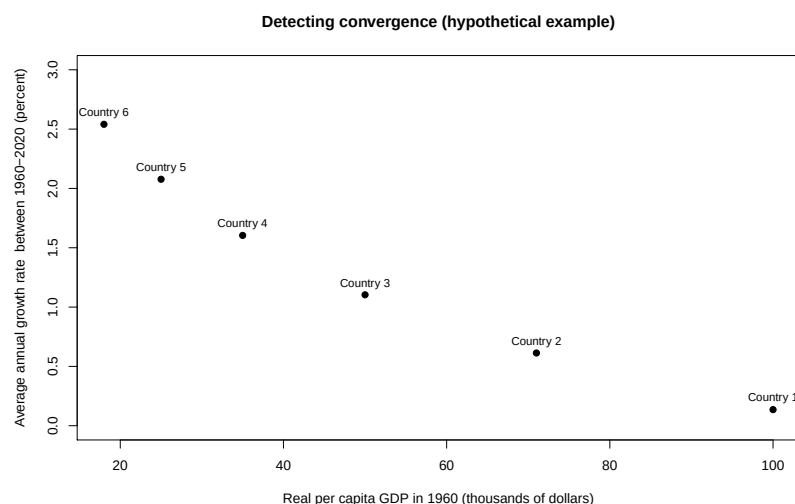
Convergence in the Data

The following graph shows the trend component of the real per capita GDP expressed in logs for some countries between 1950 and 2017 using a quadratic trend. In this group of countries, Canada (the solid black line) was the richest country in 1950 and China (the dotted green line) was the poorest. The legend includes the average annual growth rates over the period. We can see that China, Japan (dashed red line) and India (dot-dashed blue line) are converging to Canada, Nigeria (log dashed cyan line) is diverging (the gap is increasing) and Mexico (two-dashed magenta line) is neither converging nor diverging (constant gap). Therefore, convergence does not seem to be a rule for all countries.



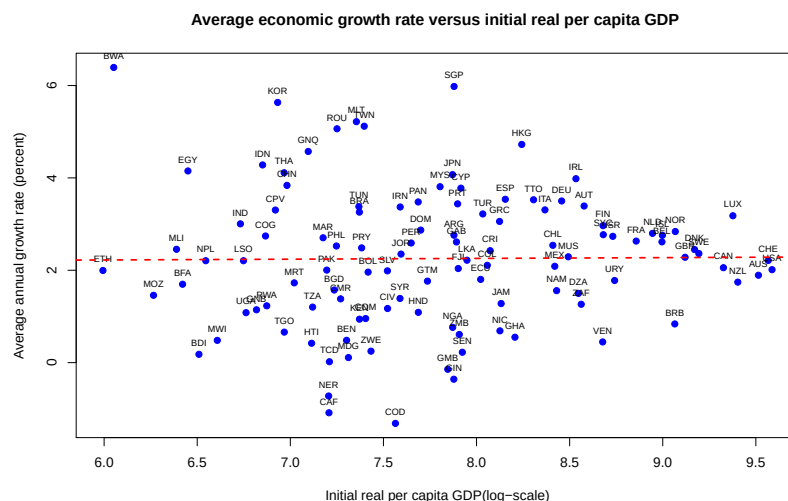
Observing Convergence using Scatter Plots

We can easily verify if there is a global tendency for the poor countries to grow faster. All we need is a scatter plot of the average annual growth rate over the whole period with respect to the initial real per capita GDP. The average growth rate can be computed by averaging the log differences ($\log(X_t) - \log(X_{t-1})$) for each country. A negative comovement between these two variables means that countries that were initially poor grow faster than countries that were initially rich, which is what convergence implies. For example, the following is the scatter plot using the hypothetical series from the previous section. The graph shows a clear negative comovement between the two variables: the richest country in 1960 (Country 1) is the one with the lowest average growth rate between 1960 and 2020 and the poorest country in 1960 (Country 6) is the one with the highest average growth rate.



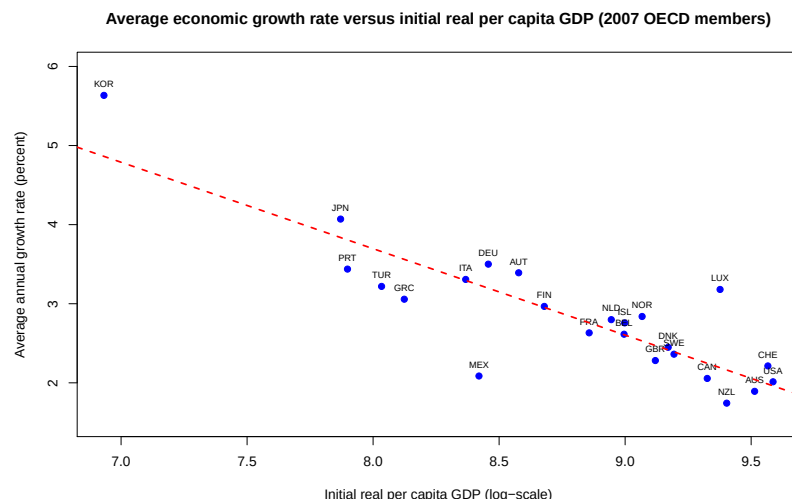
The following is the scatter plot using the 111 countries for which the initial real per capita GDP was available in 1960 and before, using the file `Characteristics.csv`. On the x-axis, we have the initial real per capita GDP (the initial year varies from country to country) and on the y-axis it is the

average growth rate between the year when the first real per capita GDP is available and 2017. The red dashed line is added to show you the average comovement between the two variables. Clearly, we do not observe overall convergence: among the initially poorest countries, the growth rates range between -1.3% and 6.4%. For example, the average economic growth rate for the Democratic Republic of the Congo (COD) from 1950 to 2017 is -1.32%. It means that its real per capita GDP was divided by more than 2 during that period ($X_T/X_0 = (1 - 0.0132)^{67} = 0.411$). On the other extreme, the average growth rate for the Republic of Korea (KOR), an equally poor country in 1953, is 5.63%: its per capita GDP was multiplied by 33.296 ($X_T/X_0 = (1 + 0.0563)^{64} = 33.296$) between 1953 and 2017. Therefore, some countries that were initially poor are converging but others are diverging.



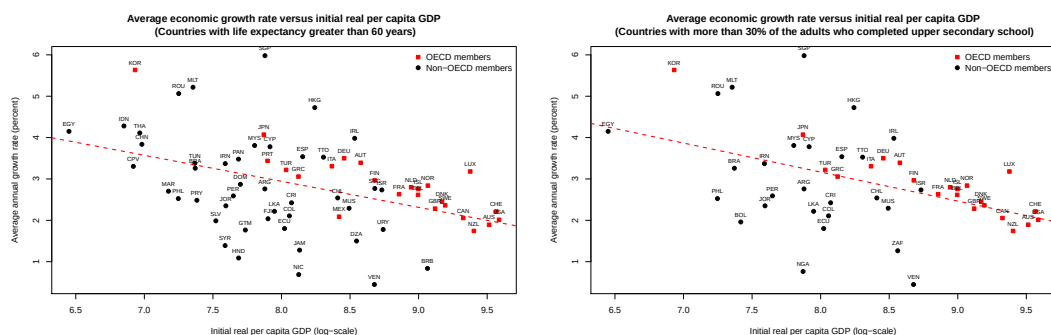
Conditional Convergence

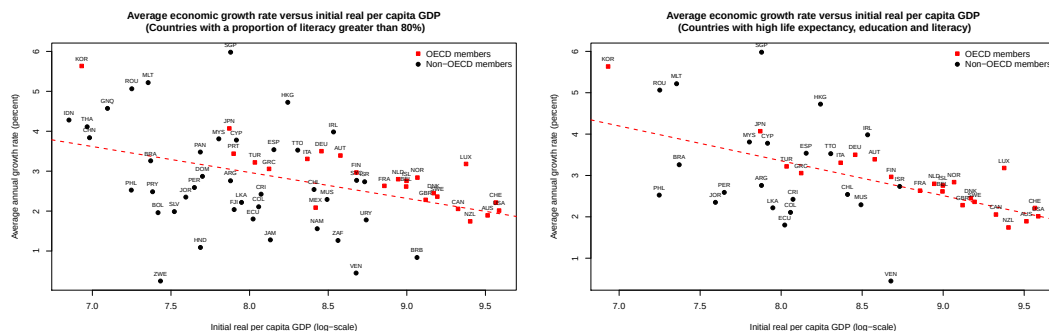
The following graph is the same scatter plot restricted to the countries that were members of the OECD in 2007 (the choice of 2007 is arbitrary). We detect a negative comovement between the variables (look at the red dashed line), which implies that OECD members are converging to each other on average. We do, however, observe some variations: Republic of Korea (KOR) and Japan (JPN) are converging much faster than Mexico (MEX) and Turkey (TUR).



The previous graph shows that convergence may happen if we select the countries wisely. The reason for selecting the countries based on their OECD membership, is that OECD countries are similar in many aspects. When we observe convergence among countries that share similar characteristics, we call it conditional convergence because the convergence is conditional on the countries having some similarities.

The economic growth theory provides a possible list of characteristics that need to be similar for countries to converge to each other. In this chapter we only explore a few: education, literacy and health. The following scatter plots include countries with the following characteristics: life expectancy greater than 60 years (top left), the education level of more than 30% of adults is at least upper secondary (top right), the proportion of literacy is greater than 80% (bottom left) and all previous three characteristics are satisfied (bottom right). The data on education, literacy and life expectancy are from Characteristics.csv file. In order to see that OECD members share those characteristics, which may explain why convergence exists among them, they are represented on the graphs by red squares. The main result is that convergence exists on average among countries with favorable characteristics (good education, good health etc.): the red dashed lines shows that the comovements are negative on average. However, the large variations suggest that being healthy and educated is not sufficient for poor countries to grow fast.

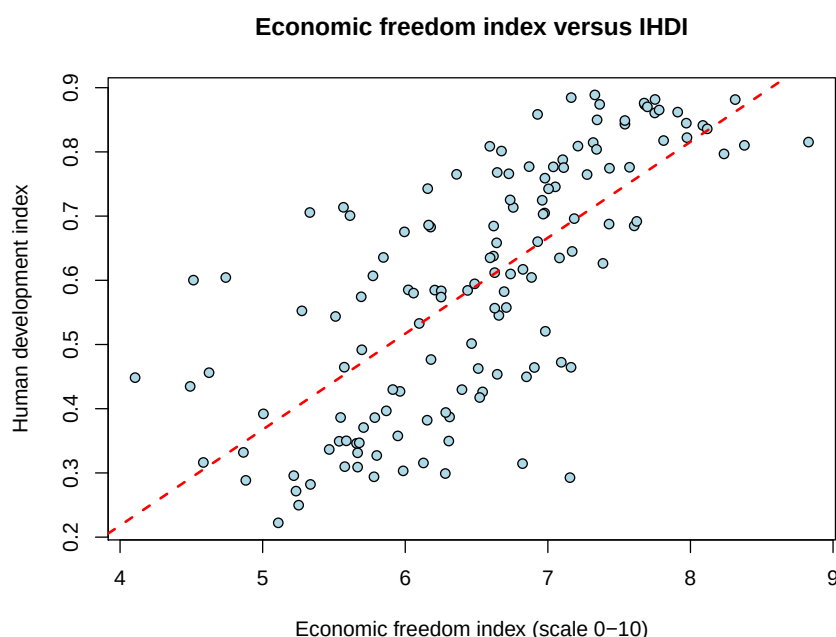




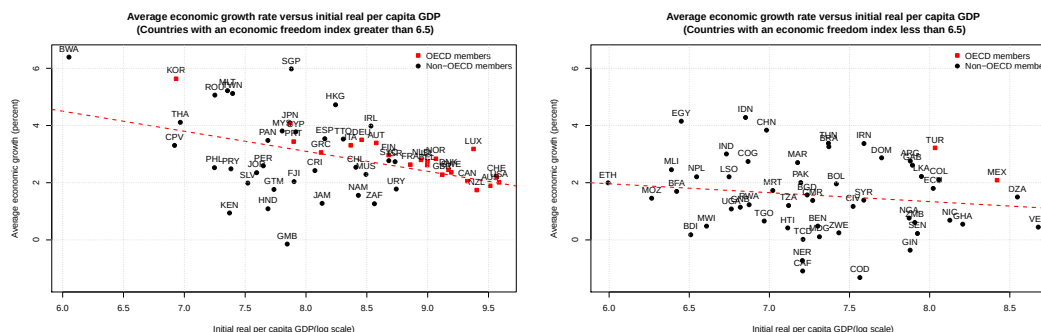
To conclude this section, we want to consider convergence conditioned on the level of economic freedom. What is economic freedom and why it may have an impact on growth and the standard of living? Watch the following short videos to understand what it means:

- Fraser Institute: The Benefits of Economic Freedom (1:51 min)
- Women's Economic Rights—What's Changed and Why Does It Matter? (1:09 min)

The Fraser Institute constructs an economic freedom index (Fraser Institute, 2020) which is based on 44 variables covering topics such as legal system, security of property rights, freedom of trade, regulation etc. The index goes from 0 (absence of freedom) to 10 (total freedom). If you are interested to know how it is constructed, you can visit their website. This is an index that puts different weights of different freedom measures. You may or may not agree with their view of what freedom means. In fact, you should always be skeptical about such measures and verify by yourself how they are constructed. An interesting results, however, is the strong positive comovement between the economic freedom index and the inequality adjusted human development index (IHDI) as we can see on the following graph. In particular, most countries with an economic freedom index greater than 7.2 have a relatively high IHDI. However, some countries have an economic freedom index around 7 and a very low IHDI.



The series of the indexes for several countries from 1970 to 2017 are included in the EconFreedom.csv file. According to the index, for example, the country with the highest economic freedom in 2017 was Hong Kong with 8.91, and Venezuela was the country with the lowest score with 2.58. The following graphs are scatter plots of economic growth with respect to initial real per capita GDP for countries with an economic freedom greater than the median 6.5 (left graph) and less than 6.5 (right graph) on average over the period 1970-2017. We observe more convergence among countries with higher economic freedom (stronger negative comovement). But again, we see lots of variations: countries with a higher economic freedom index do not necessarily converge to rich countries.

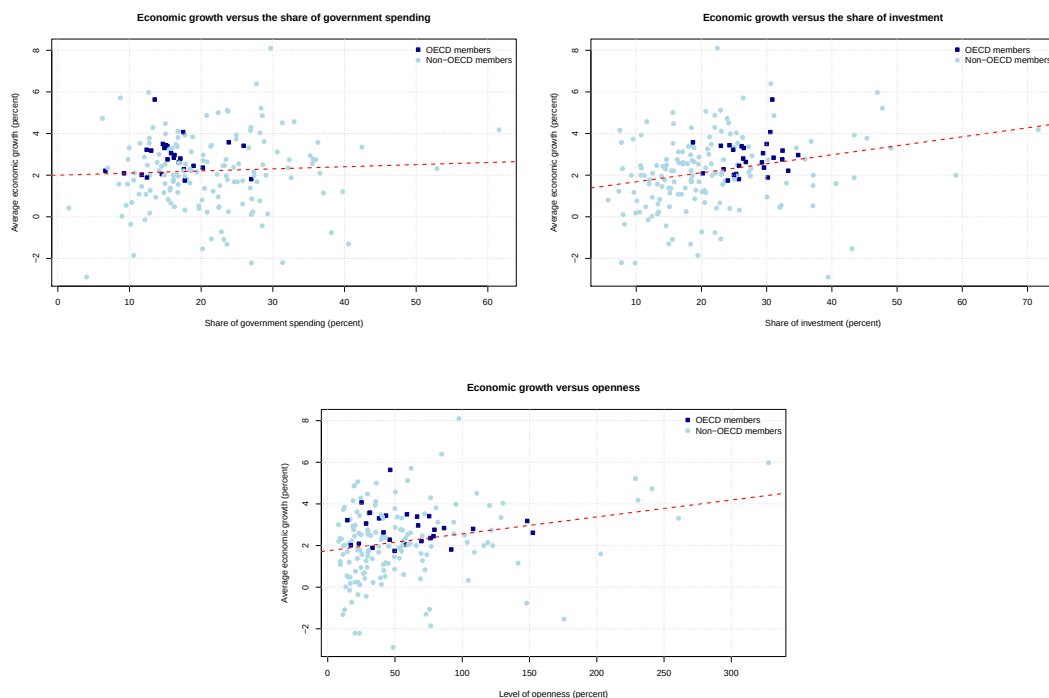


Notice that we cannot conclude that any of the variables analyzed in this section help poor countries become richer. We did observe a relationship between education, health, economic freedom and the ability of poor countries to grow, but observing comovements does not imply that an actual relationship exists. Also, the variations that we observe in all graphs suggests that we need to dig deeper to understand why some poor countries grow and others remain poor. Answering those questions is the topic of economic growth and development courses.

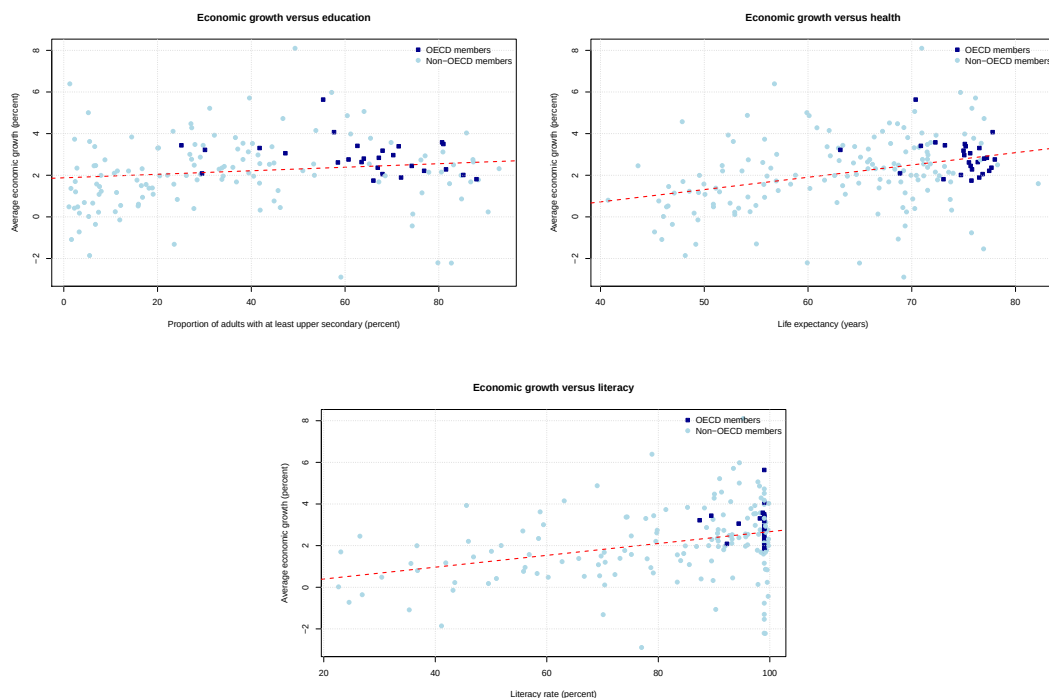
2.5 Determinants of Economic Growth

The title of the section is a little too ambitious given the tools that we have at hand. Only theory and/or advanced statistical techniques can identify what determine growth. You may find that I am repeating myself, but I want to make sure you remember that comovement and relationship are two completely different things.

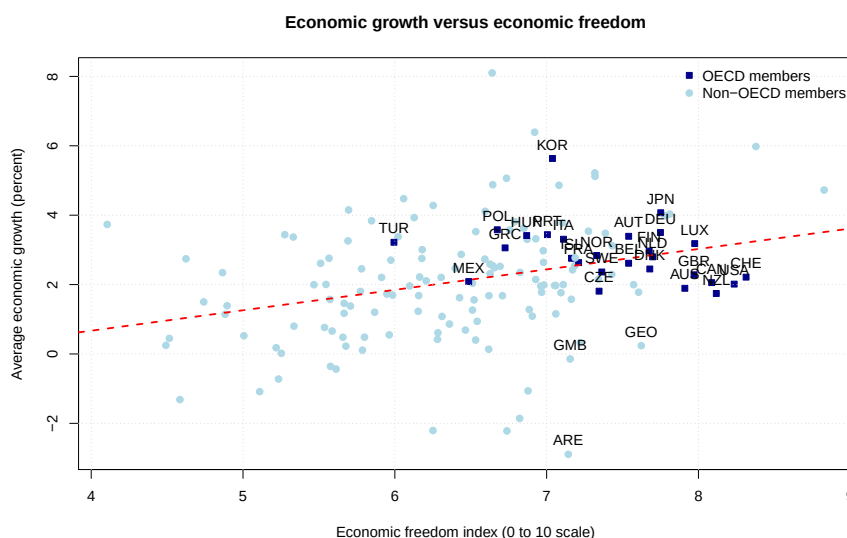
In this section we want to analyze the comovement between economic growth and some characteristics. In the following scatter plots average economic growth is on the y-axis and the country characteristics obtained from the Characteristics.csv file are on the x-axis. First, we want to see if a comovement exists between economic growth and some economic variables. The first set of variables are the government share (G divided by GDP) on the top left graph, the investment share (I divided by GDP) on the top right graph and the level of openness (X+M divided by GDP) on the bottom graph. On each graph, the OECD countries are represented by the dark blue squares and all other countries by the light blue bullets. We observe a positive comovement in all graph (shown by the red dashed line), but the it is stronger for the investment share and level of openness than for the share of government spending. Economic theory would suggest that investment and trade are good for growth. Therefore, the graphs are in line with the theory (without proving it of course).



For the next set of graphs, we want to see if we observe a comovement between economic growth and education (the top right graph), literacy (the bottom graph) and health (the top right graph). We also observe a positive comovement between those variables and the economic growth. It is less strong for the education measure, but we used a narrowed measure of education.



We conclude this section by analyzing the comovement between average economic growth and the economic freedom index. The purpose of presenting this variable is to emphasize the fact the economic growth and development is not only affected by economic variables. Characteristics such as the political system, the social norms and the legal system also play a role in a country development. The scatter plot shows that there is a positive comovement between growth and economic freedom. However, it is possible for countries with a relatively high level of economic freedom to have a low economic growth. See for example Georgia (GEO) and Gambia (GMB). The United Arab Emirates (ARE) is an outlier (a country very different from the rest) because its real per capita GDP went from 252 thousand in 1970 to 66 thousand in 2017. That can be explained by the fact that the proportion of GDP that was coming from oil production went from more than 40% in 1970 to about 10% in 2017 (source, World Bank) while the population went from half a million to 9.4 million. The Gini coefficient is not available for that country, but we should expect a high level of income inequality in 1970.



2.6 Concluding Remark

In this section, we review the main findings of the chapter. We went through many facts regarding economic growth and it is important to summarize the most important ones.

- We use real per capita GDP as an indicator of welfare not because welfare only depends on it, but because countries with high real per capita GDP tend to have high values for variables that affect the quality of life. However, we have to keep in mind that the comovement is not perfect. Many countries have high real per capita GDP and low education and life expectancy.
- The real per capita GDP has increased in most countries between 1950 and 2017, but the inequality in the growth rates resulted in an increase of income inequality over the period. If we weight the country by their population to obtain the income distribution at the individual level, the increase of income inequality is reduced. This is mainly due to the fact that large countries such as China and India have experienced relatively high economic growth.
- We do not observe unconditional convergence: globally, the standard of living of poor countries is not converging to the standard of living of rich countries.

- We do observe conditional convergence: if we restrict our sample to countries with favorable characteristics (high education, life expectancy and economic freedom), we observe poor countries converging to rich countries on average. However, there is lots of variations which means that some poor countries with favorable characteristics do not grow enough to catch-up with the richer countries.
- Economic growth seems to positively correlate (positive comovement) with investment, openness, health, education and economic freedom. However, the observed variations suggest that it is not sufficient to have high investment, openness, health, education and economic freedom to grow.
- All the facts that we analyzed in this chapter have policy implications. If we are willing to accept that growth is a solution to welfare improvement, then policies that promote economic growth are likely to be beneficial for a country. How do we promote growth? According to our results, if a poor country is able to increase its education, investment, international trade, economic freedom and health, growth should increase on average. Also, by matching the characteristics of richer countries its standard of living should converge to the standard of living of the rich countries (conditional convergence). The effectiveness of these policies relies on the validity of our observed results. Unfortunately, what we discovered in this chapter is not sufficient to assess the effectiveness of policies. In order to propose or evaluate policies, we need to know more about the theory of economic growth and development, and we need stronger statistical foundation.

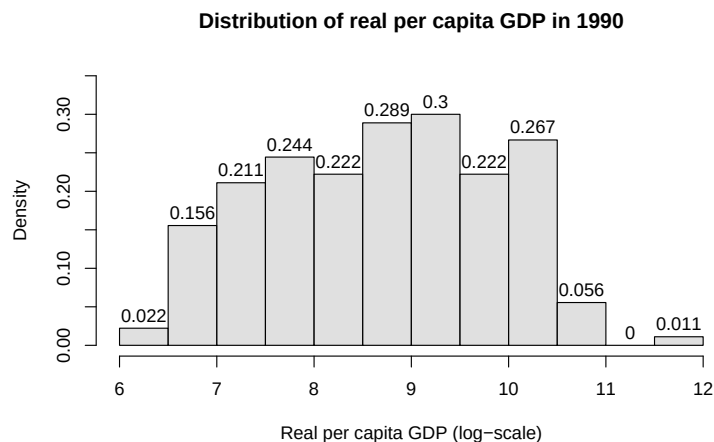
3 Exercises on Economic Growth

Question 1

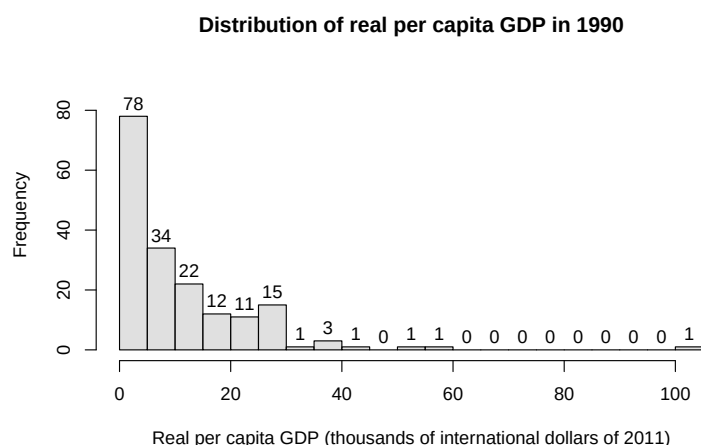
Using the file WB_LIFE.csv, compare the distribution of life expectancy across countries in 1960 and 2018 using histograms. Add the frequency on top of each bar. What is the proportion of countries with life expectancy less than 50 years in 1960 and 2018?

Question 2

- a) The following graph is the histogram of real per capita GDP in 1990 using the density as measurement unit. Given that the number of countries is equal to 180 and the interval length of each bar is 0.5, what is the proportion and the number of countries that have a log per capita real GDP between 8.5 and 9.0?



- b) The following is the same distribution with the y-axis expressed in the number of countries and the x-axis in thousands of international dollars of 2011. The number of bars has also been increased. Given that the number of countries is equal to 180 and the interval length for each bar is 5, what is the density value of the first bar? What is the main difference between the histogram in a) and b)?



- c) The following table shows the average real per capita GDP in logs and thousands of dollars per quintile.

Average real per capita GDP per quintile		
	Real per capita GDP (thousands of dollars)	Real per capita GDP (logs)
First quintile	1.14	7.00
Second quintile	2.92	7.95
Third quintile	6.39	8.74
Fourth quintile	12.86	9.44
Fifth quintile	30.20	10.24

Using the information in the table, explain why the distribution of the variable expressed in thousands of dollars is skewed while it is not (or almost not) when it is expressed in logs.

Question 3

The following is the list of countries that were members of the European Union (EU) in 2013.

Austria (AUT)	Estonia (EST)	Italy (ITA)	Portugal (PRT)
Belgium (BEL)	Finland (FIN)	Latvia (LVA)	Romania (ROU)
Bulgaria (BGR)	France (FRA)	Lithuania (LTU)	Slovakia (SVK)
Croatia (HRV)	Germany (DEU)	Luxembourg (LUX)	Slovenia (SVN)
Cyprus (CYP)	Greece (GRC)	Malta (MLT)	Spain (ESP)
Czech Republic (CZE)	Hungary (HUN)	Netherlands (NLD)	Sweden (SWE)
Denmark (DNK)	Ireland, Republic of (EIRE) (IRL)	Poland (POL)	United Kingdom (GBR)

- a) For this question, consider only the EU countries for which the 1950 real per capita GDP was available (the InitYear is 1950 in the Characteristics.csv file). Select the two countries with the lowest initial real per capita GDP and two with the highest. Then, plot the real per capita GDP of the four countries on the same chart for the period 1950-2017. Do you see poor countries converging to the richer countries?

- b) Answer question a), but express the real per capita GDP's in logs.
- c) Using all EU countries, create a scatter plot with the average growth rate on the y-axis and the initial real per capita GDP on the x-axis. Do you observe a global convergence among EU countries?
- d) Given the comovement between growth and other economic variables observed in the chapter, can you give a possible explanation of why Ireland (IRL) has grown faster than Italy (ITA) despite its initially higher real per capita GDP? How certain are you of your answer? Use the information included in the Characteristics.csv file.

Question 4

The following are the countries from West Africa. They are all included in our data files.

Benin (BEN)	Ghana (GHA)	Liberia (LBR)	Nigeria (NGA)
Burkina Faso (BFA)	Guinea (GIN)	Mali (MLI)	Senegal (SEN)
Côte d'Ivoire (CIV)	Gambia (GMB)	Mauritania (MRT)	Sierra Leone (SLE)
Cabo Verde (CPV)	Guinea-Bissau (GNB)	Niger (NER)	Togo (TGO)

- a) Select the two countries with the lowest initial real per capita GDP, two with the highest and Cabo Verde (CPV). Then, plot the real per capita GDP of the four countries on the same chart for the period available. Do you see poor countries converging to the richer countries?
- b) Answer question a), but express the real per capita GDP's in logs.
- c) Create a scatter plot with the average growth rate on the y-axis and the initial real per capita GDP on the x-axis. Do you observe a global convergence among EU countries?
- d) Given the comovement between growth and other economic variables observed in the chapter can you give a possible explanation of why the growth rate of Capo Verde is so high while it is negative for Niger (NER) and Guinea (GIN)? How certain are you of your answer? Use the information included in the Characteristics.csv and EconFreedom.csv file.

Question 5

For the following questions, you will need the following formula: let X_0 be the initial value, X_t be the value after t periods and g be the growth rate by period, then

$$X_t = X_0(1 + g)^t.$$

You may also need the log properties: $\log(a^b) = b \log(a)$ and $\log(ab) = \log(a) + \log(b)$. The properties imply:

$$\log(X_t) = \log(X_0) + t \log(1 + g).$$

- a) Suppose the initial real per capita GDP for countries A and B is 5 thousand dollars. If the annual growth rates of countries A and B are respectively 1% and 2%, what is the the ratio X_B/X_A after 60 years? Interpret the result.
- b) Suppose the annual growth rates of countries A and B are respectively 1% and 2%. How many years it will take for each country to double their respective real per capita GDP?
- c) Suppose the initial real per capita GDP of countries A, B and C are respectively 10, 10 and 50 thousand dollars. If their annual growth rates are respectively 3%, 4% and 1.5%, how many years it will take for countries A and B to converge to country C?

- d) For this question, we suppose that all countries in the PWT will growth in the future at the same rate they have grown on average from 1950 to 2017 (The Growth column in Characteristics.csv). The following table shows the real per capita GDP in 2017 for some countries and their average annual growth rate between 1950 and 2017.

	Annual growth rate (percent)	real per capita GDP in 2017 (thousands of dollars)
Canada	2.06	44.49
Nepal	2.21	2.45
Hungary	3.41	27.53
Greece	3.06	25.38
Lao People's DR	4.88	6.56

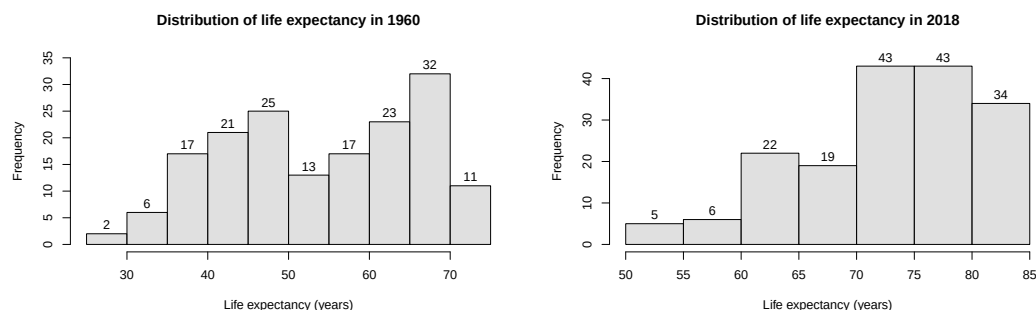
In what year each countries will reach the real per capita GDP of Canada?

- e) According to the results we obtained in the chapter, is it reasonable to assume that all countries in the PWT will growth in the future at the same rate they have grown on average from 1950 to 2017?

3.1 Solution

Question 1

The following histograms show that life expectancy has increased in general over the period. In 1960, life expectancy was below 50 years for $(2+6+17+21+25)/167 = 42.5\%$ of the countries. In 2018, no country had a life expectancy less than 50 years. In fact, life expectancy was above 70 years for the majority of the 172 countries.



Question 2

- a) The proportions are the area of the rectangles. The height of the bar associated with the interval 8.5 to 9.0 is 0.289 and its base is 0.5, so the area is $0.289 \times 0.5 = 0.1445$. Therefore 14.45% of the countries are located in that interval. Since the total number of countries is 180, the number of countries in the interval is $0.1445 \times 180 = 26$.
- b) The density times the interval length is the proportion. The proportion of countries with a real per capital GDP between 0 and 5 thousand dollars is $78/180 = 0.4333$. The density value is therefore the proportion divided by the interval length: $0.4333/5 = 0.087$. The main difference between the a) and b) histograms is that the one in b) is skewed to the right while the one in a) is not skewed.
- c) The difference between the third and first quintiles is 5.25 thousand dollars and the difference between the fifth quintile and the third is 23.81 thousand dollars. Therefore, the tail is must longer on the right of the third quintile (which is the center). Using the variable expressed in logs, the

difference between the third and first quintiles is 1.74 and the difference between the fifth quintile and the third is 1.5. Therefore, there is little difference between the length of the tail on the right and left sides of the third quintile. The reason is that differences in proportion is almost equal and differences in dollars are very different: If X_i is the average per capita GDP for the i^{th} quintile, then:

$$\log(X_3) - \log(X_1) = \log\left(\frac{X_3}{X_1}\right) = 1.74$$

$$\frac{X_3}{X_1} = e^{1.74} = 5.7$$

and

$$\log(X_5) - \log(X_3) = \log\left(\frac{X_5}{X_3}\right) = 1.5$$

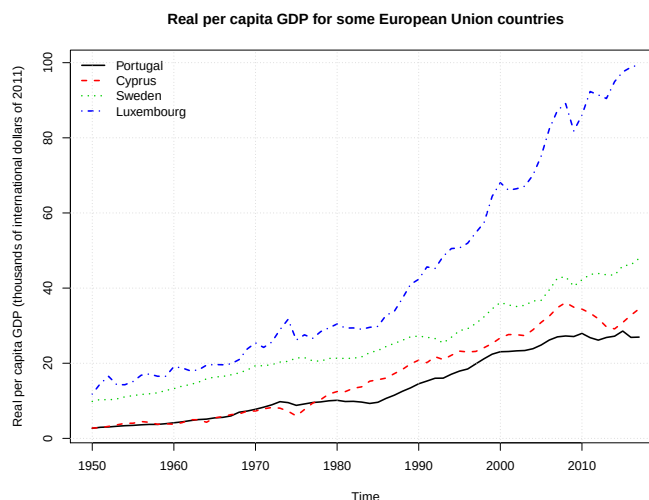
$$\frac{X_5}{X_3} = e^{1.5} = 4.48$$

Therefore, the average real per capita GDP for the third quintile is 5.7 time the one for the first quintile and the average real per capita GDP for the fifth quintile is 4.48 times the one for the third quintile. The difference is not as high as the difference in dollars.

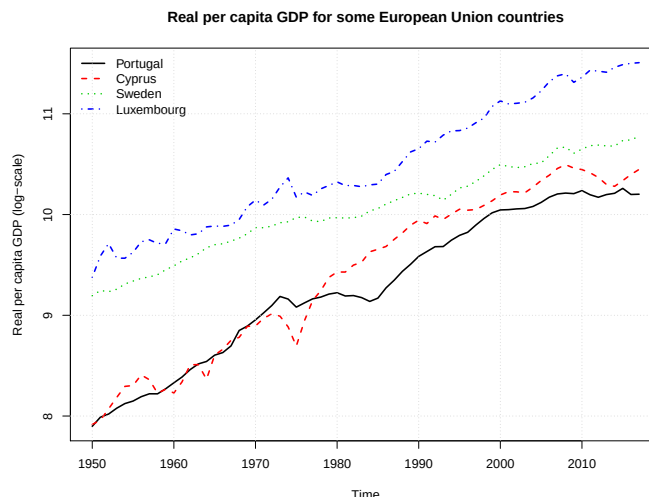
Think about it, the average income in the first quintile is 1.14 thousand. Therefore, 5.7 times 1.14 thousand means a difference of 5.358 thousand ($[5.7 \times 1.14] - 1.14$). On the other hand, the average income in the third quintile is 6.39 thousand. Therefore, 4.48 times 6.39 thousand means a difference of 22.2372 thousand.

Question 3

- a) The two initially poorest countries are Portugal (PRT) and Cyprus (CYP) and the two initially richest countries are Sweden (SWE) and Luxembourg (LUX). On the following chart, they are represented respectively by the solid black, the dashed red, the dotted green and the dot-dashed blue lines. No country is converging to the country that was initially the richest (LUX). Cyprus is getting closer to Sweden (it is converging slowly) but the gap between Portugal and Sweden has not changed much over the period (no convergence).



- b) The same line shapes and colors are used for the four countries in the following chart. The chart tells a similar story, but the gaps are now proportions which makes it a little harder to interpret.



If we compare Luxembourg and Sweden, we clearly see that gap increased in proportion. This is consistent with the huge increase in the gap in dollars that we see on the previous answer. Always remember that if $\text{gap} = \log(X_2) - \log(X_1)$, where in this case X_2 and X_1 are respectively the real per capita GDP of Luxembourg and Sweden in thousands of dollars, then

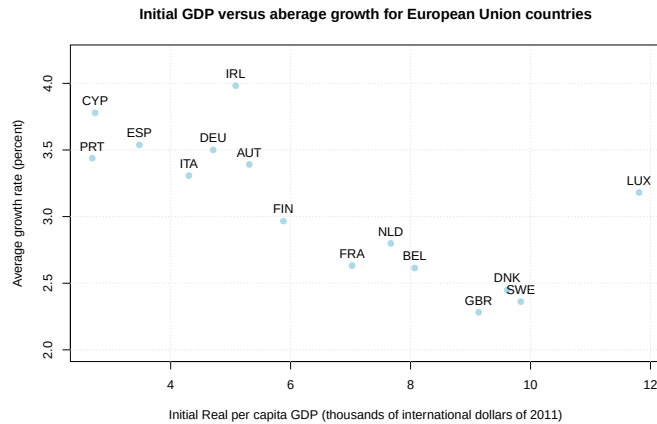
$$X_2 = X_1 e^{\text{gap}},$$

which implies the following difference (subtract X_1 on both sides):

$$X_2 - X_1 = X_1 (e^{\text{gap}} - 1).$$

Therefore, if the difference in logs is constant (same gap) but X_1 is increasing (Sweden is growing), then the gap in dollars is increasing. If we compare Portugal and Sweden, the gap in logs is decreasing over the period (the lines are getting closer to each other) but it is constant in dollars. What happens is that the increase in X_1 is offset by the smaller gap, so that the difference in dollars remains constant.

- c) The following is the scatter plot on which the country codes have been added. If we ignore Ireland (IRL) and Luxembourg (LUX) we almost get a perfect negative comovement between the two variables. Therefore, we observe convergence among EU countries.



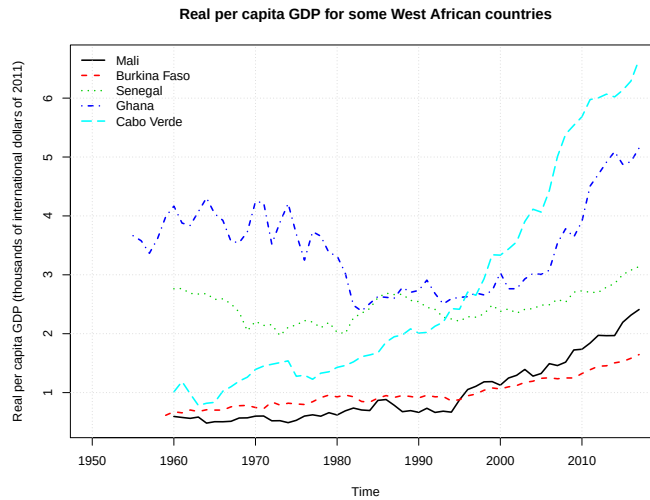
d) The following are the characteristics found in the Characteristics.csv file for Ireland and Italy:

	GovShare	InvShare	Openness	Life	Education	Literacy
Ireland	15.03	24.43	95.27	75.10	61.19	99.00
Italy	14.87	26.48	38.14	76.51	41.77	98.22

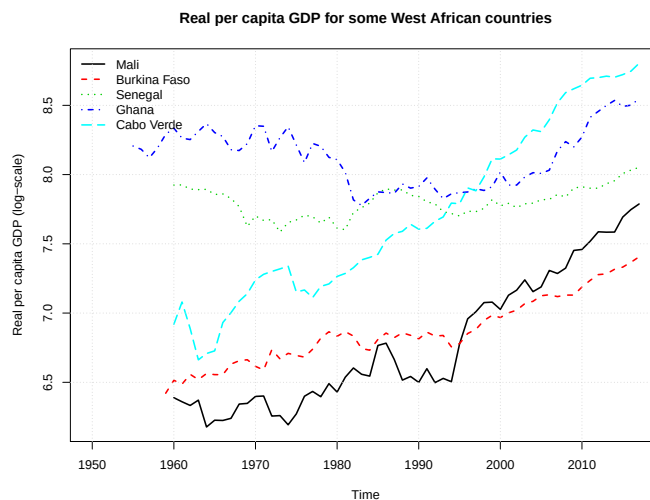
All characteristics are similar except for the education and level of openness which are much higher for Ireland. We saw in the chapter that countries with higher level of openness and larger proportion of adults with at least an upper secondary degree tend to grow faster on average. Therefore, the higher education and level of openness could explain why Ireland grew faster than Italy. Am I certain that education and openness explain the difference? The answer is no. First, we saw lots of variation in the scatter plots of growth versus characteristics: many countries with lower level of openness and education grow faster. Also, comovements do not imply direct relationships. It is just a possibility that needs to be justified by theory.

Question 4

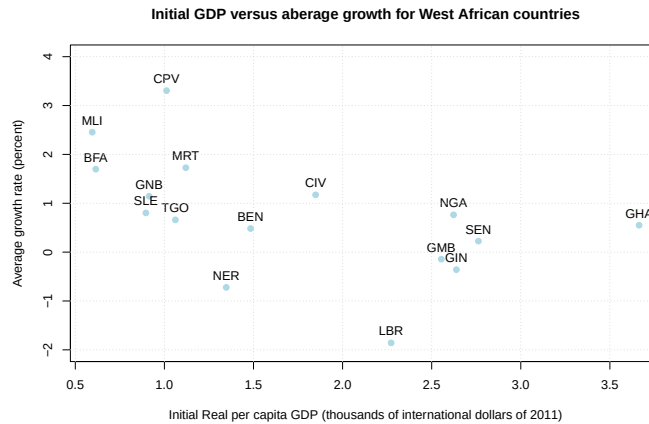
- a) The two initially poorest countries are Mali (MLI) and Burkina Faso (BFA) and the two initially richest countries are Senegal (SEN) and Ghana (GHA). On the following chart, they are represented respectively by the solid black, the dashed red, the dotted green and the dot-dashed blue lines, and Cabo Verde is the long-dashed cyan line. First, not only Cabo Verde converged to the initially richest countries, it became the richest. It is clearly the country with the highest growth rate. The Senegal and Ghana did not grow much during the period. In fact, it took 60 years for them to see their real per capita GDP's exceed their 1950 level. As a result, the initially poorest countries are converging to Senegal and Ghana despite their small (but positive) growth rate.



- b) The same line shapes and colors are used for the five countries in the following chart. The chart tells a similar story. Notice that from 2000, Cabo Verde, Ghana and Mali are growing at the same rate (the lines are parallel and we use the log-scale). But since they are different real per capita GDP, the gaps in dollars are increasing.



- c) The following is the scatter plot on which the country codes have been added. We observe a negative comovement, which implies convergence, but there are much more variation compared with the EU countries. Therefore, many countries that were poor in the '50s are still poor today (four countries have an average growth rate less than 0).



- d) The following are the characteristics found in the Characteristics.csv and EconFreedom.csv for Cabo Verde, Guinea and Niger. For the economic freedom index, we report the average over the available period.

	Growth	GovShare	InvShare	Openness	Life	Education	Literacy	EconFreedom
Cabo Verde	3.30	15.97	49.07	41.30	62.93	20.04	77.76	6.87
Guinea	-0.36	10.17	8.20	22.47	46.92	6.74	26.89	5.58
Niger	-0.72	22.83	20.05	17.83	45.21	3.28	24.54	5.23

We will ignore differences in the government spending share because we have not observed a strong comovement between this variable and the growth rate in the chapter. If we compare Cabo Verde with the other two countries, there is a clear difference. Investment share is 5 times the share in Guinea and more than twice the share in Niger. Also, there are at least 3 times more individuals with at least an upper secondary education in Cabo Verde and the literacy rate is also 3 times higher. There is also a large difference for the Economic freedom index, the level of openness and life expectancy. For the economic freedom, it is hard to interpret what it means to have a difference of 1.3. Let's say, given that it is 8.08 in Canada, that Cabo Verde is between Guinea and Canada. With these huge differences, we can say that all of these differences have something to do with the relative success of Cabo Verde. However, we need the theory to explain why they might affect growth and what are their relative importance.

For Niger and Guinea, they have similar characteristics. The investment is higher in Guinea but the openness and education are lower.

Question 5

- a) You can answer the question in two ways. First, you calculate the final real per capita GDP for each countries:

$$X_A = 5 \times (1.01)^{60} = 9.08$$

$$X_B = 5 \times (1.02)^{60} = 16.41$$

Which implies:

$$\frac{X_B}{X_A} = 1.81.$$

You can also calculate it directly:

$$\frac{X_B}{X_A} = \frac{5 \times (1.02)^{60}}{5 \times (1.01)^{60}} = \frac{(1.02)^{60}}{(1.01)^{60}} = \left(\frac{1.02}{1.01}\right)^{60} = 1.81$$

It means that real per capita GDP in country B is 1.81 times higher than in country A after 60 years.

- b) Since the real per capita GDP doubles, $X_t/X_0 = 2$. Therefore:

$$2 = (1 + g)^t$$

Knowing g , we solve for t . For country A, we have:

$$\begin{aligned} 2 &= 1.01^t \\ \log(2) &= t \log(1.01) \\ \frac{\log(2)}{\log(1.01)} &= t \\ 69.7 &= t \end{aligned}$$

Similarly, we have the following for country B:

$$t = \frac{\log(2)}{\log(1.02)} = 35$$

Therefore, it takes 35 years for country B to double its real per capita GDP and 69.7 years for country A.

- c) We are looking for t that will make the final values of two countries equal. For countries A and C, we need to solve the following equation:

$$10 \times (1.03)^t = 50 \times (1.015)^t$$

Using the properties of logarithm, we obtain:

$$\begin{aligned} \log(10) + t \log(1.03) &= \log(50) + t \log(1.015) \\ t[\log(1.03) - \log(1.015)] &= [\log(50) - \log(10)] \\ t &= \frac{\log(50) - \log(10)}{\log(1.03) - \log(1.015)} \\ t &= 109.7 \end{aligned}$$

Similarly we obtain the following for country B:

$$\begin{aligned} t &= \frac{\log(50) - \log(10)}{\log(1.04) - \log(1.015)} \\ t &= 66.1 \end{aligned}$$

Therefore, it will take 109.7 years for country A to converge to country C and it will take 66.1 years for country B.

- d) We can apply the same formula used in the previous solution. You just need to plug the numbers from the table in it. The following shows the results:

	Annual growth rate (percent)	real per capita GDP in 2017 (thousands of dollars)	Number of Years	Year of convergence
Canada	2.06	44.49		
Nepal	2.21	2.45	1974	3991
Hungary	3.41	27.53	37	2054
Greece	3.06	25.38	58	2075
Lao People's DR	4.88	6.56	70	2087

The third column is the number of years to converge to Canada (rounded to the nearest integer) and the last of the year when the country will have reached the real per capita GDP of Canada (2017+number of years). For example, it will take 1974 years for Nepal to converge to Canada. Therefore, it will happen in the year 3991.

- e) It is probably not reasonable. Even if some poor countries have not grown over the period, countries with favorable characteristics (high education, investment share, health) tend to grow at a slower rate on average as they get richer (conditional convergence). Therefore, it is not reasonable to assume that growth rate will remain constant as real per capita GDP increases.

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